Review on Virtual Synchronous Generator Model and Control for Improving Microgrid Stability

Mohd. Brado Frasetyo*
Department of Electrical and Information Engineering
Universitas Gadjah Mada
Yogyakarta 55281, Indonesia
mohd.brado41@mail.ugm.ac.id

Fransisco Danang Wijaya
Department of Electrical and Information Engineering
Universitas Gadjah Mada
Yogyakarta 55281, Indonesia
danangwijaya@ugm.ac.id

Eka Firmansyah
Department of Electrical and Information Engineering
Universitas Gadjah Mada
Yogyakarta 55281, Indonesia
eka.firmansyah@ugm.ac.id

***Abstract*—The integration of distributed generation based on renewable energy resources in microgrid systems are increasing in recent years. However, the intermittent characteristics of renewable energy result in some stability issues. One solution towards microgrid stability improvement is to provide virtual inertia by using virtual synchronous generator (VSG), which is adapting the conventional synchronous generator dynamic characteristics. Despite of the increasing VSG's development, there is no review study that observes various models and control algorithms of the technology. To fill this gap, this paper presents a review of different modeling and control algorithms that were presented along with a detailed description of parameters. This paper can be used as a guide for starting a new research on VSG model and control algorithm in microgrid applications. Finally, the main technical challenges, relevant key issues, suggested research needs, and a new perspective are discussed.**

***Index Terms*—Microgrid, Virtual Inertia, Virtual Synchronous Generator, Microgrid Stability, Modeling and Control**

I. Introduction

The integration of distributed generation based on renewable energy resources in microgrid system is increasing for recent years. In Indonesia, the estimated renewable energy installation by the end of 2028 are about 16.7 GW [1]. However, the installation of renewable energy in microgrid has several negative impacts on the system stability, such as power fluctuations and frequency instability, especially when the microgrid is in islanded mode [2]. These negative impacts are caused by renewable energy resources that have less inertia compared to conventional energy sources, whereas the inertia has an important role in maintaining the system stability.

A solution toward microgrid stability improvement is to provide additional inertia virtually. Virtual inertia is obtained by utilizing energy storage system (ESS) and power electronic along with reliable control mechanisms [3]. This concept is known as virtual synchronous machine (VISMA) or virtual synchronous generator (VSG) [4].

The basic idea of VSG with some approaches are presented in [5] [6] [7] [8]. VSG will work like a synchronous generator, which is producing a certain amount of inertia and damping characteristic. According to literature, the primary model concept are similar. However, there are some development of the model in case of adjusting with different disturbance in the microgrid. Furthermore, the control parameters have several algorithm such as classical PID controller, fuzzy logic controller (FLC), etc. Therefore, the review of VSG model and control are important to address which model and control algorithm are fit for certain microgrid application.

The currently available review on VSG possesses the basic ideas of VSG. Hasssan Bevrani et al. [9] explains the fundamentals and main concepts of VSG, the roles of VSG in microgrid, VSG topologies, VSG applications and VSG operations. Also, Khalid M. C. [10] discuss the VSG stability analysis that contains analysis of small signal stability and transient stability. For the best knowledge of the researchers, there is no such review paper that discussed the classifications of detailed model and control algorithm of VSG.

To fill this gap, this paper presents a review of several model and control algorithm of VSG that presented along with detailed description of parameters. The literature of this research has no certain time frame. However, some fresh literatures are included for the novelty of this paper. The literatures were obtained from accredited database journals and conference, such as ScienceDirect, Scientific Reports, Springer, and IEEEXplore. This paper can be used as a guide for starting a new research on VSG model and control algorithm in microgrid applications. Finally, the main technical challenges, relevant key issues, suggested research needs, and a new perspective are discussed.

II. Virtual Synchronous Generator Models

Virtual synchronous generator (VSG) is a combination of control algorithm, energy storage system and power electronic converters that emulates the characteristic of conventional synchronous generator model [10]. According to literature, the first virtual synchronous generator concepts were introduced by Beck et al. [4], well known as vitual synchronous machine (VISMA). Furthermore, some different VSG models were introduced namely virtual synchronous generators by VSYNC project, synchronverter, and VSG algebraic type. Based on the basic concept of VSG, the characteristics of each model are similar; however, the implementation of mathematical equations are slightly difference. Various model of VSG are discussed in this section.

*Corresponding author. Tel.:+62-82136803046. Address: Power System Laboratory, Dept. of Electrical and Information Engineering, Yogyakarta 55281 Indonesia.

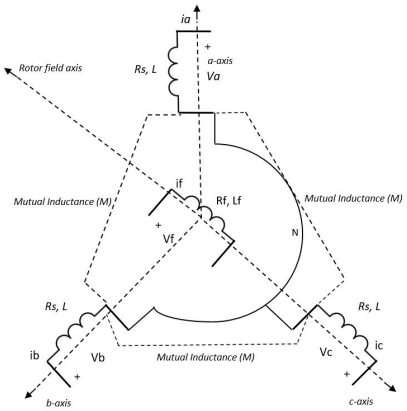


Fig. 1: Synchronous Machine Physical Winding

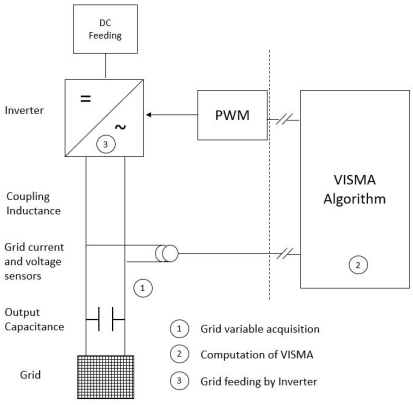


Fig. 2: Virtual Synchronous Machines (VISMA) Block Diagram

A. Virtual Synchronous Machines (VISMA)

Based on the time line of VSG's developments, VISMA is the first proposed algorithm. The VISMA algorithm is based on the combining of the static and dynamic operating properties of conventional synchronous machines with dynamic inverter technology [4]. VISMA were adopting the parameter of synchronous machine as shown in Fig.1 such as resistances (R), inductances (L), and mutual inductances (M) that specified in d-q-notation. The stator, damper and exciter are indexed in "dq", "DQ" and "e" respectively. Fig. 2 shows that VISMA has three stages of operation namely measurement, calculation/control and power injection. In the first stage, the voltage on the network will be measured and then converted into the d-q model [11].

Furthermore, the calculation of the current value of stator (i_d, i_q), damper (i_D, i_Q) and excitation (i_E) are carried out. This value is calculated in d-q frames using the following equation.

$$i_d = \frac{1}{L_d}(\Psi_d - M_{dD} i_D - M_{ed} i_e) \tag{1}$$

$$i_q = \frac{1}{L_q}(\Psi_q - M_{qQ} i_Q) \tag{2}$$

$$i_D = \frac{1}{L_D}(\Psi_D - M_{dD} i_q - M_{ed} i_e) \tag{3}$$

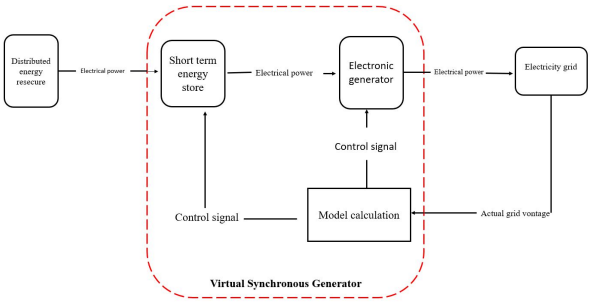


Fig. 3: Virtual Synchronous Generator (VSG) By VSYNC Project Block Diagram

$$i_Q = \frac{1}{L_Q}(\Psi_Q - M_{qQ} i_q) \tag{4}$$

$$i_e = \frac{1}{L_e}(\Psi_e - M_{eD} i_D - M_{ed} i_d) \tag{5}$$

The result of the voltage conversion and the calculated currents will be used to calculate the flux value at stator (Ψ_d, Ψ_q), damper (Ψ_D, Ψ_Q) and excitation (Ψ_e) using the following equation.

$$\Psi_d = \int (U_d - R_d i_d + \omega \Psi_q) dt \tag{6}$$

$$\Psi_q = \int (U_q - R_q i_q + \omega \Psi_d) dt \tag{7}$$

$$\Psi_D = \int (-R_d i_d) dt \tag{8}$$

$$\Psi_Q = \int (-R_q i_q) dt \tag{9}$$

$$\Psi_e = \int (U_e - R_e i_e) dt \tag{10}$$

Based on the electromechanical synchronous machine, the inner torque (m_{el}) and speed (ω) equation are as follow.

$$m_{el} = \frac{3}{2}(\Psi_d i_q - \Psi_q i_d) \tag{11}$$

$$\omega = \frac{1}{J} \int (m_{el} - m_{mech}) \tag{12}$$

The purpose of VISMA is to set the value of active and reactive power that injected from inverter. To overcome these purpose, the virtual torque (eq. 12) and voltage excitation (eq. 10) must be adjusted.

The last step of VISMA algorithm is to generate the instantaneous current value for the inverter. The previous d-q equation has to be transformed to the three phase notation.

B. Virtual Synchronous Generators by VSYNC

The next VSG model is an algorithm based on a project in western Europe that implement the virtual synchronous machines stabilizing the frequency in the distribution networks where there is a high degree of decentralized power generation (VSYNC project) that carried out by Visscher K. et al. [12]. Fig. 3 shows the block diagram of VSG that proposed by VSYNC project. The mechanical power P_{mech} that delivered to the system are based on:

$$P_{mech} = \frac{1}{2}.\frac{d}{dt}J(t).\omega(t)^2 + J(t).\omega(t).\frac{d}{dt}\omega(t) \quad (13)$$

Where the J and ω are inertia and angular frequency, respectively. Furthermore, the power injected from inverter are the reverse of the mechanical power (eq. 13). Therefore, when the angular frequency ω of the VSG are increased, the power will be injected to the energy storage which is come from the main grid. In purpose to increase the grid stability, each VSG algorithm must be act such that it eliminates grid frequency changes. Furthermore, the power injected to energy storage is equals the power absorbed from the grid, resulting in eq. 14. This model can contribute to increase the stabilization of the grid frequency. However, from the point of view of the grid side, the VSG system dynamics are considered as different from an conventional SG.

$$P_{store} = -P_{gridinjection} = P_{mech} \quad (14)$$

C. Synchronverter

Synchronverter were using the same physical winding model of VISMA as shown in Fig. 1. However, the synchronverter does not require to measure the voltages and currents reference. Furthermore, the synchronverter dynamics will be the same with SG seen from the grid side point of view. Fig. 4 shows the block diagram of synchronverter model.

Based on the block diagram, there are three main parts of synchronverter namely data input/output, dynamic model and control block diagram. The input of synchronverter model are current i, voltage v, angle θ and speed $\dot{\theta}$. Meanwhile, the output is PWM signal that sends the control signal to the inverter. Furthermore, the dynamic models of synchronverter are based on the electromagnetic torque (T_e), back electromotive force (EMF) (e) and the active/reactive power (P, Q) equation that presented in eq. 15, eq. 16 and eq. 17, respectively.

$$T_e = M_f i_f(i, sin\theta) \quad (15)$$

$$e = \dot{\theta} M_f i_f sin\theta \quad (16)$$

$$P = \dot{\theta} M_f i_f(i, sin\theta); Q = -\dot{\theta} M_f i_f(i, cos\theta) \quad (17)$$

Where M_f and i_f are the field mutual inductance and mutual inductance currents, respectively. Finally, the last part of the synchronverter is the block diagram of control. Fig. 4 shows the error of active and reactive power of VSG are sent to the dynamic equation of VSG. In the control block, there are droops coefficient of frequency D_p and voltage D_q that set based on the changes of rotor speed and voltage as follow:

$$D_p = -\frac{\Delta T}{\Delta \dot{\theta}}; D_q = -\frac{\Delta Q}{\Delta v} \quad (18)$$

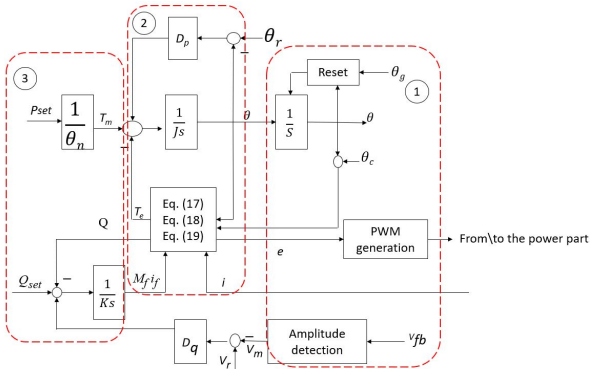


Fig. 4: Synchronverter Block Diagram

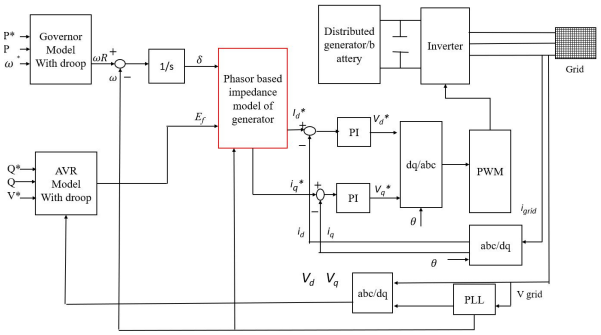


Fig. 5: Simplified VSG with Algebraic Model Block Diagram

D. VSG Algebraic Model

The VSG algebraic model is the model that the generator's steady state represented by phasor diagrams instead of Park's equations, and not to consider the generator's electrical dynamics. Furthermore, the simplified algebraic model block diagram are shown in Fig. 5.

III. VSG Model Developments

From the four basic VSG models above, there are several features that can be developed. Some researchers develop the VSG model by adding certain features that can improve the performance of the VSG algorithm. These features will be explained in the following section.

A. Power Sharing Abilities

When several VSG are integrated into one microgrid, there are some frequency and voltage deviations that caused by inherent trade of droop characteristic between voltage regulation and power sharing accuracy. One solution to overcome these problems is to manage VSG with master and slave control [5]. In this method, there are two VSG that act as slave and the master. The slave VSG is controlled as current sources and the master VSG is controlled as voltage source.

The proposed master slave VSG can provide the power sharing without any communication among slave VSG. However, the power and demand estimations are only

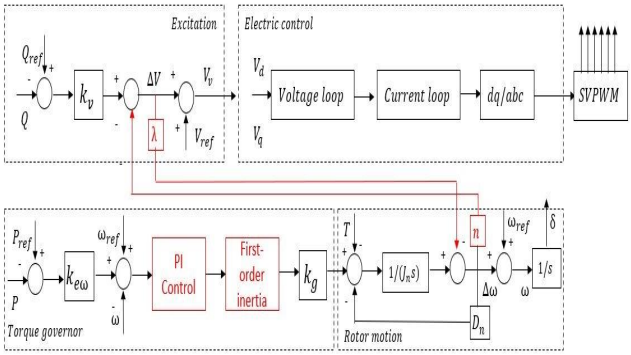


Fig. 6: VSG with Coupling Compensation Model Block Diagram

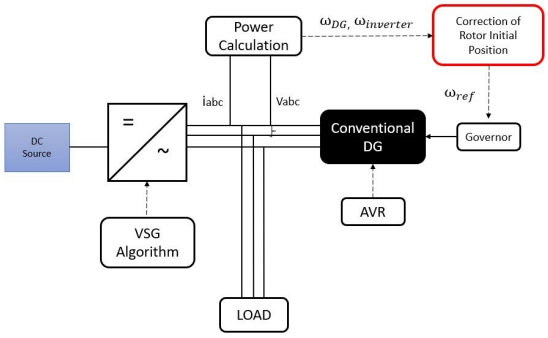


Fig. 7: VSG with Rotor Initial Position Compensation Model Block Diagram

based on addition between master and slave VSG. In addition, there must be some error due to disturbance that not included in the calculation.

B. Coupling Compensation Techniques

VSG control strategy is lack of decoupling ability, so the coupling of the DG units is uncontrollable [7] [13]. The active and reactive power value of conventional VSG is controlled base on the angular frequency and voltage value, respectively. Therefore, these types of VSG control strategies are considered as two-dimensional control, which means that the coupling of the DG units is uncontrollable. The uncontrollable coupling affects their power regulation precision, causing a bad stability in the microgrid, such as lack of dynamic performance, low power sharing accuracy. To solve these problems, a VSG control strategy with improved governor control is proposed. The control strategy is used for the coupling compensation. Its formula is written as.

$$\omega = \omega_{ref} + \frac{T_m - T}{J_n s + D_n} + \frac{\lambda J_n s(V_{ref} - V_v)}{J_n s + D_n} \tag{19}$$

$$V_v = V_{ref} + k_v(Q_{ref} - Q) + \eta(\omega_{ref} - \omega) \tag{20}$$

The parameters η and λ are coefficient of the angular frequency deviation compensation and coefficient of the voltage deviation compensation, respectively. Based on the above equation, the active and reactive power are both decided by voltage and angular frequency. Fig. 6 shows the improved VSG control strategy.

The coupling compensation can make the influence of the uncontrollable coupling of the DG units on the power regulation greatly decreased. It also can increase the power regulation degree of freedom.

C. Synchronization Abilities

Synchronization between the VSG output and the internal microgrid network or to the main network is very important. When there is a conventional synchronous generator in the same microgrid, synchronization can be done using the rotor initial position correction method [14]. The purpose of this method is to ensure the same

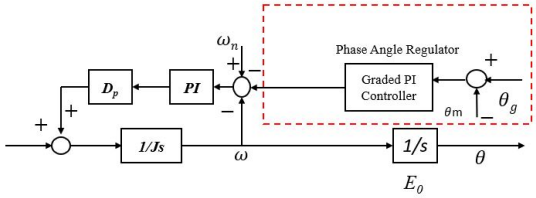


Fig. 8: VSG with Pre-Synchronization Method Block Diagram

power sharing between the two sources. So, it is sufficient to impose a zero-angle of the rotor initial position ($\alpha = 0$). As shown in Fig. 7. This correction can be achieved through an acceleration or deceleration to the DG rotor according to its initial position.

When the microgrid is switching from islanded to connected with utility grid, a significant impact may occur since the output voltage phase angle of the microgrid with the utility are different [15]. Therefore, the regulation of microgrid output with the utility is necessary. The regulation must ensure that the angle between the VSG and the main grid had a deviation that was within tolerable limits. To provide those ability, a pre-synchronization method is proposed. The process of synchronization will continue until the desired value is achieved. When the pre-synchronization process is completed, VSG can be directly connected to the main grid. Thus, the phase angle between VSG and the grid can be regulated. Fig. 8 shows the pre-synchronization control block diagram.

D. High Voltage Applications

When microgrid operates in high voltage applications, advanced converter technology is needed. One of converter technology that can be used is modular multilevel converter (MMC) [16]. MMC is a converter technology that achieves multilevel by connecting several converter in series, which has advantages compared with other

multilevel converters: the switching device withstands lower stress, less harmonic frequency and high number of the output levels, it is more flexible and convenient for capacity allocation and more suitable for different power occasions. And MMC is the main equipment of the grid-connected inverter, it can effectively improve the output efficiency, waveform, economy and reliability of the converter system, and has a wider application prospect.

E. Computing Technique and Measurement Technologies

The ability of VSG to respond to frequency changes of the grid depends on the network operation and real-time measurements of the frequency. Therefore, the needs of effective VSGs and flexible design are directed at developing monitoring/measurement technologies and computing techniques to achieve optimal performance. Most important part for the quality of VSG algorithm are the state of charge estimation and frequency estimation block. Generally, phase locked loop (PLL) is used as the frequency estimation block [17]. However, PLL has intrinsic characteristics that usually become a problem. On one hand, the voltage measurement at PCC is the critical thing for the PLL. Therefore, in case of a connection to a weak grid, the frequency and angle estimation are not ideal and are affected by the inverter behavior. Furthermore, PLL are sensitive to angle leading transients, phase jumps in the grid voltage, and frequency estimation during grid faults or transients.

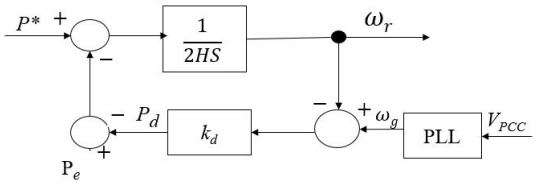
The solution for the PLL application in the VSG algorithm were introduced by Mandrile et al. [17] [18] [19] [20]. Fig. 9a shows the block diagram for conventional VSG that describing the swing equation. Meanwhile, the proposed method implements a q-axis virtual damper winding in the VSG model. This consists in modified q-axis magnetic model of the VSG and adding electrical equation for the damper [19]. The damping is not integrated into the swing equation, therefore, the value of kd is 0. Fig. 9b shows the final swing equation.

F. Smart Controller Implementations

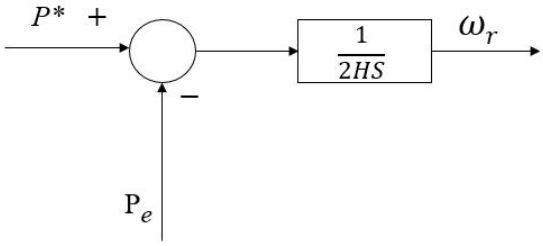
As previously described, the controller that used in VSG model is the classic PID controller. There was an experiment that using intelligent control such as a fuzzy logic controller (FLC) carried out by [6]. The enhanced VSG control with FLC is proposed as a novel control technique without any communication in a microgrid. However, the fuzzy control needs some additional requirement such as expansive number of estimations. Furthermore, the rule matrix of fuzzy are human based, so that needs to be a consideration to achieve an accurate control.

IV. VSG Problems and Future Research Directions

The development of renewable energy integration is rapidly increasing. Therefore, power converter control strategy such as VSG algorithm has been made significant progress about the last decade. Based on the model and control that aforementioned before, there are some benefits of VSG that contributes for microgrid stabilization. However, in-depth knowledge and substantial literature review show that there are many issues related to VSG integration that need to be thoroughly investigated.



(a)



(b)

Fig. 9: The PLL Reduction Diagram : (a) VSG with PLL Estimation Block; (b) VSG without PLL Estimation Block

A. Reliable Modeling

If a large number of VSG are integrated to the system, it can changes the power system crucial parameter. So, the development of reliable and detail modeling of VSG are essential to integrate the large VSG into the system without changing the parameters. On the other hand, if the VSG are totally adopting the conventional synchronous generator, the VSG will have bad properties of an SG. For example, hunting phenomena(oscillations around the synchronous frequency) as well as loss of stability due to under excitation, could occur in a VSG [21]. Therefore, a detailed, reliable, accurate and adjusted dynamic model of VSG and power system is required to overcome the trade off problems.

B. Smart Controller Developments

Generally, the VSG-based inverter uses the classic PID control system. Only a few model that adopts intelligent control systems such as fuzzy [6]. However, the VSG control system using fuzzy control is still not as good as classic PID. Therefore, in future research, the development of a VSG control algorithm using intelligent control such as artificial intelligent system are potential to improve microgrid performance. For example, feedfoward control strategy that implementing machine learning or deep learning systems for load or intermittent generation forecast. With this control strategy, the unstable generation and load that impacting the system performances will be prevented.

C. Energy Storage System Optimization

Generally, the VSG-based inverter are using batteries and capacitors as the storage systems. However, batteries and conventional capacitor have some drawbacks such as slow response. Therefore, an advanced energy storage system are required. For example, ultracapacitor based energy storage system which significantly minimises the release time of stored energy. Moreover, the combination of battery and ultracapacitor are better because they can operate at high and low frequencies. [10]. Although this solution is quite better for energy storage but not economical due to the high cost. Therefore, it is needed to develop a new and economical energy storage system which own the properties of conventional batteries and the ultracapacitor. In addition, the size of energy storage must be calculated to get the optimal size for desired microgrid system.

D. High Penetration of Renewable Sources

The present literatures show that the VSG-based inverter only supply a small/medium power to the grid (W/kW) [17] [22] [15]. Theoretically, the power delivered to the grid will be good if the renewable resources were small in comparison with the grid power capacity [21]. Therefore, any fluctuation of the renewable power generators will be overcome by the conventional generators controllers, and some of these generators will also control the overall system stability, power balance and fault ride through. However, when renewable power generators provide huge power, it will become uncontrollable. Therefore, the implementation of VSG algorithm with high penetration of renewable generation to the grid must be investigated.

V. Conclusion

Virtual synchronous generator (VSG) is a combination of control algorithm, energy storage system and power electronic converters that emulates the characteristic of conventional synchronous generator model. This paper presents a review of several model and control algorithm of VSG that presented along with detailed description of parameters. Based on the model and control that dicussed, there are some benefits of VSG that contributes for microgrid stabilization. However, in-depth knowledge and substantial literature review show that there are many issues related to VSG integration that have been investigated. Therefore, in this paper, the technical challenges, relevant key issues, suggested research needs and new perspective are discussed. This review may be used as a guideline for researcher to start a new project on VSG models and control algorithm in microgrid application.

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